Conic-ideal quotients for the A_3, B_3, and H_3 secant configurations: quadratic sheet recovery and cubic orientation in types B_3 and H_3

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Abstract

Restriction to a conic forgets how its marked points were paired into secants. For the balanced B_3 and H_3 configurations, we show that the quotient by the conic equation nevertheless recovers the two factorization sheets from quadratic products of affine evaluation functions. Within signed tensor moments, the first nonzero term is an anti-invariant cubic line that detects sheet interchange. The homogenized $2q$-point configurations are self-associated and arithmetically Gorenstein, with that signed cubic as the inverse system of an Artinian reduction. The corresponding difference spans have dimensions $3, 6, 10$ for $A_3/\mathbb{F}_5$, $B_3/\mathbb{F}_7$, and $H_3/\mathbb{F}_{11}$; in type H_3, the only missing Fischer layer is $Q\mathcal{H}_2$. The base-choice cocycle generates $H^1(\mathrm{PGL}_2(11), M_8)$, so the H_3 affine quotient has no equivariant origin; its harmonic nine-space is canonically the middle Loewy layer of the eleven-point sheet permutation module. After a chosen A_4-refinement, six incidence profiles recover the exact double-coset label in $A_4 \backslash \mathrm{PGL}_2(11)/A_5$, not in general an individual matching. Separate modular and arithmetic results identify the loss on linearization and the split–fused behavior of the three marker algebras.

Keywords. conic ideal; secant matching; Coxeter configuration; self-associated points; arithmetically Gorenstein; cubic inverse system; modular representation.

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1 Introduction

The companion rigidity paper isolates the Clebsch hexagon through its uncovered syndrome locus and reconstructs its projective geometry from decoding data [7]. Here we begin with marked secant geometry and ask a complementary reconstruction question: what does common restriction to a conic forget, and what does the conic-ideal quotient remember?

The answer is governed by one mechanism. A four-endpoint switch shows that competing secant products differ by the conic equation, so their quotients form an affine configuration. In the balanced cases, Hadamard squares recover an unordered bipartition of that configuration; the first signed tensor that survives is cubic and orients the two parts. A selected double-coset incidence map then compresses the H_3 configuration to six profiles without identifying any labels.

type	linear quotient	sheets/cubic	H_3 refinements	arithmetic
$A_3/\mathbb{F}_5$	rank 3	not applicable	not applicable	fused
$B_3/\mathbb{F}_7$	rank 6	recovered/oriented	not used	split
$H_3/\mathbb{F}_{11}$	rank 10	recovered/oriented	six labels/depth	split/glued

The quotient construction and recovery mechanisms are symbolic. The A_3/B_3 rank attainments and the recovery hypotheses are exact certified inputs. For H_3, Sylow-normalizer cohomology proves the missing-layer statement and top nonvanishing; one exact scalar supplies the radial line. The profile and modular results use further finite incidence data.

Theorem 1.1 (Quadratic recovery and cubic orientation)**.** *Let the A_3, B_3, and H_3 matching configurations be the finite matching orbits defined in Section 3. Then:*

(i) *canonically scaled matching products have equal restriction to the conic, and their conic-ideal quotients have difference ranks* 3, 6, *and* 10*;*

(ii) *in the B_3 and H_3 cases, the two factorization sheets are, up to interchange, the unique complementary halves with equal first and second tensor moments;*

(iii) *the signed moments vanish through degree two, while a nonzero degree-three tensor transforms by the character that exchanges the sheets;*

(iv) *the homogenized B_3 and H_3 configurations are self-associated with Gorenstein coordinate rings; the signed cubic is the Macaulay inverse system of their Artinian reductions;*

(v) *in the H_3 case, six explicit incidence profiles separate the six A_4–A_5 double-coset labels, and the singleton profiles recover the corresponding decorated matching-table row;*

The modular depth quotient and the arithmetic split–fused theorem are stated separately in Sections 6 and 7; they refine the same finite table but are not clauses of the recovery theorem.

The exceptional H_3 configuration arose from a normal-play finite-projective arc game and subsequently led to coding and quantum applications. Its game-theoretic origin and decoding rigidity are developed in the companion rigidity paper [7]; its MDS–AME and transversal-Clifford aspects are treated separately in [8], using the standard arc–MDS–AME dictionary [11]. Prescribed-hole arcs are developed in [9]. Beyond redundancy four, the same syndrome-as-projective-geometry viewpoint is developed through Hankel witness systems and coherent polar recognition [10]. None of those results is a hypothesis here.

2 Conic matching products

Write the standard conic as

$$\mathcal{C} = V(Q), \qquad Q = XZ - Y^2,$$

and parametrize it by $\nu(s:t) = (s^2 : st : t^2)$. For points $a = (a_0 : a_1)$ and $b = (b_0 : b_1)$ of $\mathbb{P}^1$, put $[a, b] = a_0 b_1 - a_1 b_0$. We scale the secant L_{ab} through $\nu(a)$ and $\nu(b)$ by the condition

$$L_{ab}(\nu(s:t)) = [a, (s:t)]\,[b, (s:t)].$$

Let Ω be a set of $2m$ distinct marked points of $\mathbb{P}^1$, with fixed vector representatives. For a perfect matching M of Ω, define its matching product

$$P_M = \prod_{\{a,b\}\in M} L_{ab}.$$

The vector representatives are part of the marked-conic data. Their role is to make the products comparable before passing to projective equivalence.

Proposition 2.1 (The matching-secant quotient)**.** *For every $m \geq 2$ and every two perfect matchings M, N of Ω,*

$$P_M|_{\mathcal{C}} = P_N|_{\mathcal{C}}, \qquad P_M - P_N \in (Q).$$

Consequently, after choosing a reference matching M_0,

$$\Phi_{M_0}(M) = \frac{P_M - P_{M_0}}{Q} \in H^0(\mathbb{P}^2, \mathcal{O}(m-2)) \tag{2.1}$$

is an affine configuration. Changing the reference matching translates this configuration. If $g \in \mathrm{GL}_2$ preserves Ω, $\rho(g)$ is its action on binary quadratics, and representatives are transported by g, then

$$\Phi_{gM_0}(gM) \circ \rho(g) = \det(g)^{2m-2} \Phi_{M_0}(M). \tag{2.2}$$

Thus the difference space, its rank, and its affine moment conditions do not depend on M_0 and are equivariant under the endpoint stabilizer.

Proof. For $x \in \mathbb{P}^1$,

$$P_M(\nu(x)) = \prod_{a \in \Omega} [a, x],$$

which does not depend on the pairing. The kernel of the Veronese pullback

$$k[X, Y, Z] \longrightarrow k[s, t], \qquad (X, Y, Z) \longmapsto (s^2, st, t^2)$$

is the principal ideal (Q), so Q divides every difference. Changing the reference follows by subtraction. Finally, $L_{ga,gb} \circ \rho(g) = \det(g)^2 L_{ab}$ and $Q \circ \rho(g) = \det(g)^2 Q$, which gives (2.2). □

2.1 The four-endpoint switch

For four distinct endpoints, the Plücker relation gives the local switch identity

$$L_{ab}L_{cd} - L_{ac}L_{bd} = [a, d][b, c]\, Q. \tag{3.1}$$

Multiplying (3.1) by the secant factors common to two matchings proves divisibility for a single switch. Any two perfect matchings are joined by a sequence of such switches: if M and N differ at a, write $\{a, b\} \in M$ and $\{a, c\} \in N$; switching the two M-edges through b and c creates $\{a, c\}$ and reduces the symmetric difference. This also gives an elementary proof of the divisibility clause in Proposition 2.1.

The switch calculation and the global pullback proof play different roles. The pullback proves the quotient for arbitrary matchings at once. The switch identity records how local changes of factorization move in the quotient space. A selected Coxeter orbit need not contain every perfect matching or every intermediate switch.

3 The rank-three configurations

We now select three finite matching orbits. This selection is discrete input: Proposition 2.1 applies to every matching, but it does not distinguish the Coxeter orbits. Maximum matchings arise more generally as equality objects in prescribed-hole defect theory, where concurrent secants form a matching design and, dually, a star–matching incidence realization [9]. Away from equality, the same defect controls both the edge count and the secant vertex-cover number of the graph formed by nonmaximum concurrence cliques. This supplies an incidence-geometric route toward the maximum-matching regime. The comparison is combinatorial: here the matched endpoints lie on the conic, and the Coxeter orbits are additional finite geometric input.

For $q \in \{5, 7, 11\}$, index

$$\mathbb{P}^1(\mathbb{F}_q) = \{0, 1, \ldots, q-1, \infty\},$$

where a denotes $(1 : a)$ and ∞ denotes $(0 : 1)$. Let $G_q = \mathrm{PGL}_2(q)$ act in the usual way. The three base matchings are

$$\begin{array}{ccl} T & q & M_T \\ \hline A_3 & 5 & \{0,\infty\}, \{1,4\}, \{2,3\}, \\ B_3 & 7 & \{0,2\}, \{1,4\}, \{3,\infty\}, \{5,6\}, \\ H_3 & 11 & \{0,1\}, \{2,5\}, \{3,7\}, \{4,9\}, \{6,8\}, \{10,\infty\}. \end{array} \tag{4.1}$$

Define the *rank-three matching configuration*

$$\Omega_T = G_q \cdot M_T.$$

The stabilizers of the three displayed matchings are respectively S_4, S_4, A_5, and hence

$$|\Omega_{A_3}| = 5, \qquad |\Omega_{B_3}| = 14, \qquad |\Omega_{H_3}| = 22. \tag{4.2}$$

These marker spaces and their stabilizers are classical. In particular, Edge and Dye identify the exceptional finite conic configurations and substantial parts of their incidence geometry [2, 3]. We use the Coxeter names for the corresponding tetrahedral, octahedral, and icosahedral data; neither the orbit counts in (4.2) nor the ambiguity of an unmarked conic is asserted here as new.

Fix M_T as reference and apply (2.1) to every member of Ω_T. Write $R_d = \mathbb{F}_q[X,Y,Z]_d$ and

$$\Delta_Q = 4\partial_X\partial_Z - \partial_Y^2, \qquad \mathcal{H}_d = \ker(\Delta_Q : R_d \longrightarrow R_{d-2}). \tag{4.3}$$

The operator Δ_Q is the Laplacian attached to the dual quadratic form. In the degrees used below, $\mathcal{H}_d$ is the top harmonic summand, while powers of Q give radial summands.

Theorem 3.1 (The $3, 6, 10$ quotient ranks)**.** *For the matching configurations in (4.1), let*

$$W_T = \mathrm{span}_{\mathbb{F}_q}\{\Phi_{M_T}(M) : M \in \Omega_T\}.$$

Then

$$\begin{array}{cccc} T & \deg\Phi & W_T & \dim W_T \\ \hline A_3 & 1 & R_1 = \mathcal{H}_1 & 3, \\ B_3 & 2 & R_2 = \mathcal{H}_2 \oplus \mathbb{F}_7 Q & 6, \\ H_3 & 4 & \mathcal{H}_4 \oplus \mathbb{F}_{11} Q^2 & 10. \end{array} \tag{4.4}$$

If $h_T = 4, 6, 10$ is the Coxeter number and $d = h_T/2 - 1$, the three rows have the uniform form

$$W_T = \begin{cases} \mathcal{H}_d, & d \text{ odd}, \\ \mathcal{H}_d \oplus \mathbb{F}_q Q^{d/2}, & d \text{ even}, \end{cases} \qquad \dim W_T = h_T - 1 + \mathbf{1}_{\{d \text{ even}\}}.$$

In particular, the first two configurations fill the complete quotient form space. The H_3 configuration spans a canonical ten-dimensional subspace of the fifteen-dimensional space R_4.

Proof mode. The quotient construction and the harmonic-space dimensions are proved symbolically. The A_3 and B_3 spanning assertions are certificate-checked exact calculations. For H_3, a Sylow-normalizer cohomology argument excludes the middle Fischer layer and forces the top harmonic summand; one displayed outer-sheet quotient certifies the remaining radial line. A separate implementation reconstructs all three original row reductions and the smaller H_3 witness from the data in (4.1).

Proof. A matching in (4.1) has $m = 3, 4, 6$ edges, respectively, so Proposition 2.1 places its quotient in R_{m-2}, of degrees $1, 2, 4$. Representing the quotients in the standard monomial bases gives matrices with respectively $5, 14, 22$ rows. Exact row reduction over $\mathbb{F}_5$ and $\mathbb{F}_7$ gives ranks 3 and 6. Direct differentiation gives

$$\dim \mathcal{H}_1 = 3, \qquad \dim \mathcal{H}_2 = 5, \qquad \dim \mathcal{H}_4 = 9.$$

The radial vector $Q \in R_2$ is not harmonic in characteristic 7. The first two exact quotient matrices lie in $\mathcal{H}_1$ and $\mathcal{H}_2 + \mathbb{F}_7 Q$; their ranks equal the dimensions of these spaces, proving the first two rows.

We prove the H_3 row equivariantly. Put $G = \mathrm{PGL}_2(11)$, $G^+ = \mathrm{PSL}_2(11)$, and let $\chi(g) = \det(g)^5$ be the square-class character. On the conic, the common restriction of the matching products is the Dickson form $s^{11}t - st^{11}$, of contragredient weight $\det^{-1}$. Let $V = \mathbb{F}_{11}^2$, with binary forms carrying the contragredient $\mathrm{GL}_2(11)$-action. The equivariant Fischer decomposition obtained from the Clebsch identification $\mathrm{Sym}^2 V \simeq \mathbb{F}_{11}^3$, together with the Dickson weight in the quotient transformation (2.2), gives the linearized module

$$\widetilde{R}_4 = R_4 \otimes \overset{-1}{\det} = M_8 \oplus M_4 \oplus M_0, \tag{4.5}$$

Explicitly, $g \in \mathrm{GL}_2(11)$ acts on the first expression by

$$g \cdot f = \det(g)^{-1}(f \circ \rho(g)^{-1}).$$

The three summands are

$$\begin{aligned} M_8 &= \mathcal{H}_4 \otimes \overset{-1}{\det} \simeq \mathrm{Sym}^8(V^\vee) \otimes \overset{-1}{\det}, \\ M_4 &= Q\mathcal{H}_2 \otimes \overset{-1}{\det} \simeq \mathrm{Sym}^4(V^\vee) \otimes \overset{-3}{\det}, \\ M_0 &= \mathbb{F}_{11}Q^2 \otimes \overset{-1}{\det} \simeq \overset{-5}{\det} = \chi. \end{aligned} \tag{4.6}$$

Equivalently, all three summands have the uniform form

$$M_r = \mathrm{Sym}^r(V^\vee) \otimes \overset{r/2}{\det} \otimes \chi, \qquad r = 8, 4, 0.$$

A scalar λI acts here as $\lambda^{-r}\lambda^r(\lambda^2)^5 = 1$, so these modules and their direct sum descend from $\mathrm{GL}_2(11)$ to G. This is the G-action used below. The determinant twist changes only the linearization, not the underlying Fischer subspaces $\mathcal{H}_4, Q\mathcal{H}_2, \mathbb{F}_{11}Q^2$ of R_4.

We need two elementary cohomology facts. Let $U = \langle u \rangle$ be the upper-unipotent Sylow 11-subgroup and let a generate the diagonal torus of order 10. Since $r < 10$, the cyclic norm $1 + u + \cdots + u^{10} = (u-1)^{10}$ vanishes on M_r, while $(u-1)M_r$ has codimension one. Thus $H^1(U, M_r)$ is represented by the class of x^r. Diagonal conjugation acts on that class by

$$a^{4-r/2}. \tag{4.7}$$

Restriction from G to U is injective because $[G:U]=120$ is invertible in $\mathbb{F}_{11}$, and its image is fixed by the Sylow normalizer. Equation (4.7) therefore gives

$$H^1(G,M_4)=H^1(G,M_0)=0, \qquad \dim H^1(G,M_8)\le 1. \tag{4.8}$$

The stabilizer $H=A_5$ of M_{H_3} lies in G^+. The restrictions of M_8, M_4, M_0 to H are respectively $4\oplus 5, 5, 1$, so

$$M_8^H=M_4^H=0, \qquad \dim M_0^H=1. \tag{4.9}$$

Equivalently, average the characters $(9,1,0,-1,-1)$, $(5,1,-1,0,0)$, and $(1,1,1,1,1)$ over the A_5 class sizes $(1,15,20,12,12)$.

Let $c(g)=\Phi_{M_{H_3}}(gM_{H_3})$. The common-restriction identity gives

$$c(gh)=gc(h)+c(g),$$

and c vanishes on H. Project c through (4.5). By (4.8), its M_4-component is a coboundary; its translating vector lies in $M_4^H=0$, so that component vanishes. This proves, without quotient coordinates,

$$\Delta_Q\Phi_{M_{H_3}}(M)\in \mathbb{F}_{11}Q \quad \text{for every } M\in\Omega_{H_3}. \tag{4.10}$$

Likewise the M_0-component is a coboundary and hence vanishes on G^+. Choose any other matching in the eleven-element G^+-orbit of M_{H_3}. Its quotient is nonzero: equality of the two secant products would identify their irreducible line factors and hence the matchings. Its M_4- and M_0-components vanish, so it gives a nonzero vector in M_8. The degree-eight restricted $\mathrm{SL}_2(\mathbb{F}_{11})$-module is irreducible [12, Fact 3.1]. Moreover the span of the cocycle values is G-stable, since

$$b\,c(g)=c(bg)-c(b) \qquad (b,g\in G).$$

Therefore the quotient span contains all of M_8.

It remains only to see the radial line. With ∞ indexed by 11, take

$$M_-=\{\{0,1\},\{2,\infty\},\{3,8\},\{4,6\},\{5,9\},\{7,10\}\}.$$

The nonsquare-determinant projectivity represented on V by

$$\begin{pmatrix}0&1\\1&1\end{pmatrix}$$

sends M_{H_3} to M_-; hence M_- belongs to the outer sheet of Ω_{H_3}. The primary certificate and independent replay also reconstruct this orbit membership. Exact division and differentiation give

$$\Delta_Q^2\Phi_{M_{H_3}}(M_-)=10=\Delta_Q^2(Q^2)\neq 0. \tag{4.11}$$

The independent replay reconstructs this single scalar from the secant formula. Since Δ_Q^2 kills $M_8\oplus M_4$, (4.11) supplies the M_0-projection. The span already contains M_8, so it also contains M_0. Hence $W_{H_3}=\mathcal{H}_4\oplus\mathbb{F}_{11}Q^2$, of dimension 10. Since $\dim R_4=15$, this equality is proper. □

Corollary 3.2 (The base-choice class and its homogeneous extension)**.** *Let $c_8: G\to M_8$ be the top-harmonic projection of the base-choice cocycle c. Then*

$$H^1(G,M_8)=\mathbb{F}_{11}[c_8]. \tag{4.11a}$$

In particular, the affine H_3 quotient configuration has no G-equivariant origin: no translation makes its affine action linear. Equivalently, the homogeneous module

$$E_c = M_8 \oplus \mathbb{F}_{11}, \qquad g \cdot (v,t) = (gv + t\, c_8(g), t), \tag{4.11b}$$

is nonsplit. Up to equivalence and nonzero rescaling of the quotient coordinate, it is the unique nonsplit extension of the trivial G-module by M_8.

Proof. Equation (4.8) gives $\dim H^1(G, M_8) \le 1$. On the base sheet the M_0-component vanishes, and the M_4-component vanishes by (4.9); the nonzero same-sheet quotient used in the proof above is therefore a value of c_8. If $c_8(g) = gv - v$, its vanishing on $H = \mathrm{Stab}_G(M_{H_3})$ would put v in $M_8^H = 0$, forcing $c_8 = 0$, a contradiction. Thus $[c_8] \ne 0$, proving (4.11a).

Changing the reference quotient translates the affine configuration and changes c_8 by a coboundary. Hence a G-equivariant origin would kill $[c_8]$, which is impossible. Finally the standard identification $\mathrm{Ext}^1_{\mathbb{F}_{11}G}(\mathbb{F}_{11}, M_8) \simeq H^1(G, M_8)$ identifies (4.11b) with $[c_8]$; the one-dimensionality just proved gives the last assertion. □

Corollary 3.3 (The exact H_3 loss)**.** *Over $\mathbb{F}_{11}$, the quartic form space has the Fischer decomposition*

$$R_4 = \mathcal{H}_4 \oplus Q\mathcal{H}_2 \oplus \mathbb{F}_{11}Q^2. \tag{4.12}$$

Consequently,

$$R_4 = W_{H_3} \oplus Q\mathcal{H}_2, \qquad R_4/W_{H_3} \simeq Q\mathcal{H}_2. \tag{4.13}$$

Equivalently,

$$W_{H_3} = \{f \in R_4 : \Delta_Q f \in \mathbb{F}_{11}Q\}. \tag{4.14}$$

Thus the H_3 quotient configuration retains the top harmonic nine-space and the deepest radial line, and omits precisely the five-dimensional middle radial layer.

Proof. For a homogeneous form f of degree r, direct differentiation gives

$$\Delta_Q(Qf) = (4r+6)f + Q\Delta_Q f. \tag{4.15}$$

Hence $\Delta_Q(Qh) = 3h$ for $h \in \mathcal{H}_2$, while $\Delta_Q(Q^2) = 9Q$ over $\mathbb{F}_{11}$. Applying Δ_Q^2 and then Δ_Q shows that the three summands in (4.12) meet trivially. Their dimensions are $9, 5, 1$, so they exhaust the fifteen-dimensional space R_4. Theorem 3.1 then gives (4.13). If $f = h_4 + Qh_2 + cQ^2$ is its decomposition, then

$$\Delta_Q f = 3h_2 + 9cQ.$$

The quadratic decomposition $R_2 = \mathcal{H}_2 \oplus \mathbb{F}_{11}Q$ therefore gives (4.14). □

The rank proof has two distinct boundaries. The common restriction and quotient are general consequences of the conic ideal, and (4.3) describes the intrinsic target subspaces. The A_3 and B_3 spanning steps remain full-orbit row reductions. For H_3, (4.7)–(4.10) prove the containment and top-summand nonvanishing equivariantly; only the single radial scalar (4.11) remains an exact coordinate witness. The primary certificate still enumerates every orbit and recovers the original ranks, while an independent replay reconstructs both that calculation and the smaller witness from (4.1).

This proof is uniform at the level of its mechanism—an affine connecting cocycle, its Sylow-normalizer weights, stabilizer fixed spaces, and irreducible summands—but it still uses the H_3-specific $A_5 < \mathrm{PGL}_2(11)$ module data. A fully uniform theorem for all three rank-three types would also replace the remaining A_3 and B_3 row reductions. A theorem over all good reductions would additionally require integral models and an analysis of the characteristics in which the harmonic decomposition, orbit structure, or marker algebra degenerates.

Corollary 3.3 identifies the linear information lost by the H_3 orbit, but linear span does not recover the two factorization sheets inside the finite configuration. Their recovery begins one categorical level higher, with products of affine evaluation functions.

4 Balanced sheets and cubic orientation

For $T = B_3, H_3$, put $q = 7, 11$, respectively. The subgroup

$$G^+ = \mathrm{PSL}_2(q) \subset G = \mathrm{PGL}_2(q)$$

splits Ω_T into two orbits $\Omega_+ \sqcup \Omega_-$, each of size q; an element of the outer coset exchanges them. These are the two one-factorization sheets. The A_3 orbit does not split: its S_4 stabilizer cannot lie in $\mathrm{PSL}_2(5) \simeq A_5$, already by order.

Write $x_M = \Phi_{M_T}(M) \in W_T$. Let $L \subseteq \mathbb{F}_q^{\Omega_T}$ be the evaluation space of affine-linear functions on the finite configuration $\{x_M\}$. Since the points affinely span W_T,

$$\dim L = 1 + \dim W_T = q.$$

For subspaces of functions, write

$$L^{\circ 2} = \mathrm{span}\{fg : f, g \in L\},$$

where multiplication is pointwise. The common second-moment form in the proposition below is

$$B(\lambda, \mu) = \sum_{M \in \Omega_+} \lambda(x_M)\mu(x_M) = \sum_{M \in \Omega_-} \lambda(x_M)\mu(x_M)$$

on $W_T^\vee$. A *field-valued strength-two trade* is a weight $\tau \in k^\Omega$ satisfying, for all $f, g \in L$,

$$\sum_{M \in \Omega} \tau(M) f(M) g(M) = 0.$$

Proposition 4.1 (Radical–Hadamard recovery)**.** *Let k be a field and $\Omega = \Omega_+ \sqcup \Omega_-$, with both sheets of size $q = 0$ in k. Suppose an affine evaluation space $L \subseteq k^\Omega$ has the following properties:*

(i) *each sheet has zero first moment, and the two sheets have equal second moments;*

(ii) *$\dim L = q$, and restriction from L onto the zero-sum hyperplane of either sheet has rank $q - 1$;*

(iii) *the second-moment form on linear functionals has rank $q - 2$ and a one-dimensional radical;*

(iv) *a nonzero radical functional is constant on each sheet, with different constants on the two sheets.*

Then

$$L^{\circ 2} = \left\{ (u_+, u_-) : \sum_{\Omega_+} u_+ = \sum_{\Omega_-} u_- \right\}, \tag{5.1}$$

and

$$(L^{\circ 2})^\perp = k(\mathbf{1}_{\Omega_+} - \mathbf{1}_{\Omega_-}). \tag{5.2}$$

Consequently every field-valued strength-two trade is a scalar multiple of the sheet sign.

Proof. Let e_+ and e_- be the two sheet indicators. The radical functional evaluates as $a_+e_+ + a_-e_-$ with $a_+ \neq a_-$. Together with $1 = e_+ + e_- \in L$, it therefore yields $e_+, e_- \in L$.

Let $H_\pm$ be the zero-sum hyperplane on $\Omega_\pm$. The first-moment hypothesis and $q = 0$ put each restriction of L inside $H_\pm$; the rank hypothesis makes both restrictions surjective. Multiplication by e_+ and e_- now gives

$$(H_+ \oplus 0) + (0 \oplus H_-) \subseteq L^{\circ 2}. \tag{5.3}$$

This subspace has dimension $2q - 2$.

For $f, g \in L$, the difference between the two sheet sums of fg expands into constant, first-moment, and second-moment terms. The hypotheses make all three differences zero, so $L^{\circ 2}$ lies in the hyperplane on the right of (5.1). Finally choose restrictions $a, b \in H_+$ with nonzero standard pairing and lift them to $f, g \in L$. Equality of the second moments gives the same nonzero pairing on Ω_-. Thus fg has equal nonzero sheet sums and does not lie in (5.3). We have found $2q - 1$ independent directions in the $2q - 1$-dimensional hyperplane, proving (5.1). Its annihilator under the standard coordinate pairing is the line in (5.2). ☐

The hypotheses of Proposition 4.1 are small finite linear-algebra statements for the two quotient configurations:

T	q	sheet restriction ranks	second-moment rank	$\dim L^{\circ 2}$
B_3	7	$6, 6$	5	13,
H_3	11	$10, 10$	9	21.

(5.4)

In each row the radical is one-dimensional and takes two distinct sheet values. The exact certificate checks these hypotheses from the matching quotients; an independent implementation reconstructs the row reductions and the resulting trade line. No enumeration of equal-size subsets is used in the proof.

Theorem 4.2 (Balanced sheets and cubic-first orientation)**.** *For $T = B_3, H_3$, the unordered pair $\{\Omega_+, \Omega_-\}$ is intrinsic to the affine quotient configuration:*

(i) *up to interchange, Ω_+ and Ω_- are the only complementary halves with equal first and second tensor moments;*

(ii) *if ϵ is $+1$ on Ω_+ and -1 on Ω_-, then*

$$\mu_j = \sum_{M \in \Omega_T} \epsilon(M) x_M^{\odot j} \in \mathrm{Sym}^j(W_T)$$

satisfies

$$\mu_1 = 0, \qquad \mu_2 = 0, \qquad \mu_3 \neq 0; \tag{5.5}$$

(iii) *μ_3 is independent of the reference matching, G^+ fixes it, and the outer coset negates it. Within G,*

$$\mathrm{Stab}_G(\mu_3) = G^+, \qquad \mathrm{Stab}_G(\mathbb{F}_q\mu_3) = G. \tag{5.6}$$

Thus degree three is the first signed tensor moment, or linear functional of such a moment, that orients the recovered sheet pair. This minimality statement concerns signed tensor moments, not every possible nonlinear statistic of the configuration.

Proof mode. Proposition 4.1 is a symbolic proof. The ranks, radical separation, lower-moment cancellation, and nonzero cubic for the two displayed configurations are certificate-checked over $\mathbb{F}_7$ and $\mathbb{F}_{11}$ and independently replayed. The transformation law and reference independence below are formal consequences of those finite inputs.

Proof. Equality of first and second tensor moments for a complementary pair is equivalent, by polarization, to its sign weight annihilating $L^{\circ 2}$. Since the characteristics are 7 and 11, polarization through degree two is valid. Proposition 4.1 says that the annihilator is exactly the sheet-sign line. A sign weight taking only the values $\{\pm 1\}$ is therefore ϵ or $-\epsilon$, proving (i).

The same moment equalities give $\mu_1 = \mu_2 = 0$. Exact tensor summation in the rank-six and rank-ten quotient coordinates gives $\mu_3 \neq 0$. If the reference matching changes, every point translates by one vector c. Expanding the signed third moment after $x_M \mapsto x_M + c$ leaves μ_3 unchanged: the other terms contain μ_2, μ_1, or $\mu_0 = \sum_M \epsilon(M) = 0$.

Every element of G^+ preserves both sheets and hence fixes the signed sum. Every element of the outer coset exchanges the sheets and negates it. Since $\mu_3 \neq 0$, these two transformation laws give (5.6). Finally (5.5) proves the stated minimality inside the signed tensor-moment family. □

Corollary 4.3 (Graded evaluation algebra)**.** *Put $L^{\circ 0} = \langle 1 \rangle$, and let $L^{\circ d}$ be the span of the pointwise products of d elements of L. For $T = B_3, H_3$, the pointwise powers of the affine evaluation space satisfy*

$$\dim L^{\circ d} = \begin{cases} 1, & d = 0, \\ q, & d = 1, \\ 2q - 1, & d = 2, \\ 2q, & d \geq 3. \end{cases}$$

In particular, $L^{\circ 3} = \mathbb{F}_q^{\Omega_T}$. The homogenized configuration

$$\widehat{X}_T = \{[1 : x_M] : M \in \Omega_T\} \subseteq \mathbb{P}^{q-1}$$

therefore has Hilbert function

$$1,\ q,\ 2q - 1,\ 2q,\ 2q, \ldots$$

and h-vector $(1, q-1, q-1, 1)$.

Proof. Pair functions on Ω_T with the sheet sign by $\langle \epsilon, f \rangle = \sum_M \epsilon(M) f(M)$. Proposition 4.1 gives $L^{\circ 2} = \ker\langle \epsilon, - \rangle$. Since $\operatorname{char} \mathbb{F}_q$ is 7 or 11, polarization in degree three identifies $\operatorname{Sym}^3(W_T^*)$ with the dual of $\operatorname{Sym}^3(W_T)$. The nonzero tensor μ_3 therefore pairs nontrivially with some cubic in the linear coordinates on W_T. Evaluating that cubic on the points x_M gives an element of $L^{\circ 3}$ whose pairing with ϵ is nonzero.

Because $1 \in L$, we have $L^{\circ 2} \subseteq L^{\circ 3}$. Thus $L^{\circ 3}$ contains the hyperplane $\ker\langle \epsilon, - \rangle$ and a vector outside it, so $L^{\circ 3} = \mathbb{F}_q^{\Omega_T}$. Multiplication by 1 gives the same equality in every higher degree. Homogeneous degree-d forms on $\widehat{X}_T$ restrict to $L^{\circ d}$, which gives the Hilbert function; its first difference is the stated h-vector. Changing the reference matching translates every x_M. Such a translation preserves L and all its pointwise powers and acts projectively on $\widehat{X}_T$, so the graded algebra and Hilbert function are independent of the reference. □

Corollary 4.4 (Self-association and cubic duality)**.** *For $T = B_3, H_3$, the configuration $\widehat{X}_T \subseteq \mathbb{P}^{q-1}$ is a reduced self-associated arithmetically Gorenstein set of $2q$ points. Every one-point deletion imposes independent conditions on quadrics, so $\widehat{X}_T$ has the Cayley–Bacharach property in degree two. Its dualizing residue vector is the sheet sign ϵ. The Artinian reduction by the homogenizing coordinate has Hilbert function $(1, q-1, q-1, 1)$, and its Macaulay inverse system is the cubic line $\mathbb{F}_q \mu_3$.*

Proof. Let A be the q-by-$2q$ matrix whose M-th column is $\widehat{x}_M = (1, x_M)$, and put $D = \operatorname{diag}(\epsilon(M))$. Equal sheet sizes and the vanishing first and second signed moments give

$$ADA^{\mathsf{T}} = 0.$$

The affine spanning statement gives $\operatorname{rank} A = q$. Consequently the rows of AD form the kernel of A, so AD is a Gale matrix for $\widehat{X}_T$. Its columns differ from those of A only by the nonzero scalars $\epsilon(M)$; hence the labeled configuration is its own Gale transform.

Corollary 4.3 says that the $2q$ points fail by exactly one to impose independent conditions on quadrics. The unique dependence is ϵ, which has full support, so deleting any coordinate removes that dependence and leaves rank $2q-1$. After base change to an algebraic closure, the hypotheses of Eisenbud–Popescu's self-association criterion [5, Theorem 7.3] hold. It follows that the homogeneous coordinate ring is Gorenstein after base change, and hence over $\mathbb{F}_q$.

Let z be the homogenizing coordinate. It is nonzero at every point and is therefore regular on the coordinate ring. The degree d part of the reduction modulo z is $L^{\circ d}/L^{\circ(d-1)}$, so its Hilbert function is the h-vector from Corollary 4.3. The functional

$$f \longmapsto \sum_M \epsilon(M) f(x_M)$$

annihilates degrees below three and is nonzero in degree three. Thus ϵ is the socle residue vector. Degree-three polarization identifies its Macaulay dual generator with μ_3. ☐

The implication above is the zero-defect case of the modern self-dual-code criterion: the Gorenstein defect is measured by the defect of the Schur square, or equivalently by decomposability [6]. Here $\dim L^{\circ 2} = 2q-1$ already places both configurations in the zero-defect case. The ideal, deletion, socle, inverse-system, and minimal-resolution calculations are also checked directly in the verification bundle. Self-association does not select an affine origin. Translating all x_M acts projectively on the columns of A and preserves the signed Gale identity, in agreement with the nontrivial base-choice obstruction behind the quotient construction.

The cubic remembers an orientation that the linear quotient does not. Indeed, the signed linear relation

$$\sum_{M\in\Omega_+} x_M = \sum_{M\in\Omega_-} x_M$$

already shows that the sheet sign is killed by the linear quotient map. Quadratic products recover the unordered partition through (5.2), and the cubic supplies the sign exchanged when the two recovered levels are swapped.

Corollary 4.5 (Secant-product tensor syzygies)**.** *For $T = B_3, H_3$, let $m_Q : R_{m-2} \to R_m$ denote multiplication by Q. Regarding the displayed powers as symmetric tensors, not polynomial powers, the canonically scaled secant products satisfy*

$$\begin{aligned}
&\sum_M \epsilon(M) P_M = 0,\\
&\sum_M \epsilon(M) P_M^{\odot 2} = 0,\\
&\sum_M \epsilon(M) P_M^{\odot 3} = \operatorname{Sym}^3(m_Q)(\mu_3) \neq 0.
\end{aligned} \tag{5.7}$$

Proof. Substitute $P_M = P_{M_T} + Qx_M$ and expand. Each sheet has $q=0$ elements in $\mathbb{F}_q$, and the signed quotient moments vanish through degree two. All terms below cubic degree cancel, leaving $\mathrm{Sym}^3(m_Q)(\mu_3)$ in degree three. Since $\mathbb{F}_q[X,Y,Z]$ is a domain, multiplication by Q is injective; over a field, the symmetric powers of an injective linear map are injective. $\square$

5 Six profiles and matching-row reconstruction

We now specialize to H_3. Put

$$G = \mathrm{PGL}_2(11), \qquad G^+ = \mathrm{PSL}_2(11), \qquad X = \Omega_{H_3} \simeq G/H,$$

where $H \simeq A_5$ is the stabilizer of the base matching in (4.1). Thus $X = X_+ \sqcup X_-$, with $|X_+| = |X_-| = 11$. The construction has four stages:

$$(A_+, A_-) \longmapsto K \longmapsto \text{four oriented cell pairs} \longmapsto D(M) \longmapsto K\backslash G/H.$$

The ordered parent pair supplies the A_4-subgroup K; its cell partition supplies four signed incidence counts; those counts recover the double-coset label. Only the first two stages require coordinates.

5.1 The decorated parent correspondence

On $V = \mathbb{F}_{11}^3$, set

$$q_0(x,y,z) = x^2 + y^2 + z^2, \qquad \mathcal{C}_0 = \mathbb{P}\{q_0 = 0\}.$$

The projectivity

$$\psi = \begin{pmatrix} 10 & 2 & 1 \\ 1 & 2 & 10 \\ 3 & 0 & 3 \end{pmatrix} \quad \text{satisfies} \quad q_0(\psi(X,Y,Z)^t) = 3(XZ - Y^2), \tag{6.1}$$

so $\psi \circ \nu$ identifies the endpoint conic of Section 2 with $\mathcal{C}_0$. In these coordinates take the two six-point sets

$$\begin{aligned} A_+ &= \{(0:1:4), (0:1:7), (1:0:3), (1:0:8), (1:4:0), (1:7:0)\}, \\ A_- &= \{(0:1:3), (0:1:8), (1:0:4), (1:0:7), (1:3:0), (1:8:0)\}. \end{aligned} \tag{6.2}$$

Let $\mathcal{P} = G \cdot A_+$, where G is viewed as the full projective stabilizer of $\mathcal{C}_0$. For $A \in \mathcal{P}$ and $a \in A$, let $C_{A,a}$ be the conic through $A \setminus \{a\}$. The six pairs

$$C_{A,a} \cap \mathcal{C}_0, \qquad a \in A, \tag{6.3}$$

form a perfect matching M_A of the twelve points of $\mathcal{C}_0$. On this 22-point orbit, $A \mapsto M_A$ is a G-equivariant bijection onto X. This is the *matching-decorated parent correspondence*; its bijectivity and the assertion that (6.3) consists of six disjoint pairs are exact finite inputs. It is the secant-factorization counterpart of the self-polar matching and Brianchon-support reconstruction obtained from the Clebsch decoding geometry [7].

Put $H_\pm = \mathrm{Stab}_G(A_\pm)$ and

$$K = H_+ \cap H_-, \qquad J = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}. \tag{6.4}$$

Then $K \simeq A_4$, $J(A_+) = A_-$, and J normalizes K and exchanges X_+ with X_-. Thus the *ordered golden pair* means precisely the ordered pair (A_+, A_-).

Choose linear representatives of K preserving q_0, and let $\widetilde{K} = \mathbb{F}_{11}^{\times} K$. For the following eight vectors, define $R_i = \mathbb{P}(\widetilde{K} r_i)$:

$$\begin{array}{cccc} i & r_i & i & r_i \\ \hline 1 & (0,1,4) & 10 & (0,1,3) \\ 3 & (0,1,2) & 13 & (0,1,5) \\ 6 & (1,3,2) & 14 & (1,1,5) \\ 9 & (1,4,2) & 11 & (1,2,4). \end{array} \tag{6.5}$$

These are cells in the scalar-K orbit partition of $\mathbb{P}(V)$. The involution J exchanges the four pairs

$$(R_1, R_{10}), \quad (R_3, R_{13}), \quad (R_6, R_{14}), \quad (R_9, R_{11}). \tag{6.6}$$

Only these four oriented pairs enter the argument. Reversing the golden pair reverses all four orientations.

This profile layer uses more decoration than the balanced-sheet theorem. The quotient configuration intrinsically recovers the unordered sheets, but it does not select an ordered golden pair or the resulting scalar-K refinement. The reconstruction below is relative to that selected refinement.

5.2 Incidence profiles

For $M \in X$, let S_M be the union of its six secants in $\mathbb{P}^2(\mathbb{F}_{11})$, and put

$$Z_M(R) = |S_M \cap R|.$$

Equivalently, $Z_M(R)$ counts the points of R at which P_M vanishes. Define the *depth profile*

$$\begin{aligned} D(M) = (&Z_M(R_1) - Z_M(R_{10}),\ Z_M(R_3) - Z_M(R_{13}), \\ &Z_M(R_6) - Z_M(R_{14}),\ Z_M(R_9) - Z_M(R_{11})) \in \mathbb{F}_{11}^4. \end{aligned} \tag{6.7}$$

This definition uses only zero sets of secant products, so it is unchanged if their equations are rescaled.

Theorem 5.1 (Six-profile reconstruction)**.** *The K-orbits on each sheet $X_\pm$ have sizes $1, 4, 6$. With the orientation in (6.6), the corresponding profiles and fibre sizes are*

$$\begin{array}{ccc} \textit{sheet} & |D^{-1}(v)| & v \\ \hline + & 1 & v_1 = (-6, 0, 12, -12), \\ + & 4 & v_2 = (-3, 3, 0, 3), \\ + & 6 & v_3 = (3, -2, -2, 0), \\ - & 1 & -v_1, \\ - & 4 & -v_2, \\ - & 6 & -v_3. \end{array} \tag{6.8}$$

The six vectors are pairwise distinct. Hence D recovers the double-coset row in

$$K \backslash G / H$$

of every matching. Its linear span nevertheless has dimension only two: over $\mathbb{F}_{11}$ it is the plane

$$2a + 2b + c = 0, \qquad 9a + 8b + d = 0. \tag{6.9}$$

The two singleton profiles recover their individual matchings. Under the matching-decorated parent correspondence, they therefore recover the corresponding individual H_3 parent. A profile in a fibre of size four or six recovers only that K-orbit, not an individual matching.

Proof mode. The subgroup-mark reduction and the implications from equivariance are conceptual. The six representative incidence rows in (6.10) below are certificate-checked exact counts and are reconstructed by an independent implementation. A companion certificate verifies that the matching decoration uniquely recovers its parent on the 22-point H_3 locus.

Proof. The permutation character of K on either eleven-point sheet has values

$$(11, 3, 2)$$

on elements of orders $1, 2, 3$. The marks of the transitive A_4-sets are

K-set	size	marks on orders $1, 2, 3$
K/K	1	$(1, 1, 1)$
K/V_4	3	$(3, 3, 0)$
K/C_3	4	$(4, 0, 1)$
K/C_2	6	$(6, 2, 0)$
$K/1$	12	$(12, 0, 0)$.

Comparing marks gives the unique nonnegative decomposition

$$X_+|_K \simeq K/K \ \sqcup\ K/C_3 \ \sqcup\ K/C_2.$$

The orbit sizes are therefore $1, 4, 6$; conjugation by J gives the same decomposition on X_-.

For $k \in K$, the transformation k preserves every cell in (6.6) and carries S_M to S_{kM}. Thus $D(kM) = D(M)$. The involution J exchanges the two cells in every oriented pair and carries S_M to S_{JM}, so

$$D(JM) = -D(M).$$

It remains to count one representative on each positive-sheet orbit. The exact incidence rows are

orbit size	$\mathrm{Stab}_K(M)$	zero counts in the four pairs of (6.6)	$D(M)$
1	A_4	$(0, 6;\ 0, 0;\ 12, 0;\ 0, 12)$	v_1,
4	C_3	$(3, 6;\ 3, 0;\ 3, 3;\ 6, 3)$	v_2,
6	C_2	$(3, 0;\ 1, 3;\ 2, 4;\ 6, 6)$	v_3.

(6.10)

The J-images supply the three negative rows. These six vectors are pairwise distinct, so their fibres are precisely the six K-orbits.

Row reduction over $\mathbb{F}_{11}$ gives rank two and the equations (6.9). This rank drop does not merge orbit labels: linear rank and set-theoretic separation are different assertions. Finally, a singleton fibre contains its matching without residual choice. The stabilizer of the matching M_A from (6.3) is the same A_5 as the stabilizer of its parent A. The equivariant decorated transform between the two G/H-orbits is therefore a bijection, so the matching also determines its parent. The stated limitation for the other four rows follows from their fibre sizes. ☐

Corollary 5.2 (Decorated sheet classifier)**.** *Relative to the orientation in (6.6), the profile $D(M)$ determines the sheet containing M. The antipodal singleton pair $\{v_1, -v_1\}$ recovers the unordered golden matching pair and hence the unordered golden parent pair; choosing the orientation selects one member.*

Proof. The positive and negative profile sets in (6.8) are disjoint, and the only singleton fibres are v_1 and $-v_1$. Apply the matching–parent bijection from Theorem 5.1. ☐

Perfect-matching incidence also survives in the associated six-party state: companion work converts simultaneous concurrence in the arc and its Gale dual into a local-unitary marginal-moment invariant and uses it to separate the H_3 state from the GRS locus [8]. That invariant is a different quotient: it forgets an A_5 orientation, whereas the cubic here detects the factorization-sheet character.

This pointwise classifier does not contradict the failure of a linear sheet sign in Section 4. The map D uses zero incidences with the selected scalar-K refinement; it is not a linear functional on the affine quotient configuration. Without that extra decoration, the quadratic and cubic moment statements remain the intrinsic recovery mechanism.

The profile compression also retains the cubic-first orientation from Section 4. The three positive vectors satisfy the integral weighted relation

$$v_1 + 4v_2 + 6v_3 = 0. \tag{6.11}$$

Corollary 5.3 (Orbit weights from normalized profiles)**.** *The three incidence-normalized vectors v_1, v_2, v_3 recover the orbit-size multiset $\{1, 4, 6\}$. They also recover the stabilizer orders $\{12, 3, 2\}$ inside $K \simeq A_4$. The underlying rational rays alone do not recover these weights.*

Proof. The coordinates of each v_i are fixed by the cardinality differences in (6.7), so independent rescaling is not allowed. These three integral vectors span a plane over $\mathbb{Q}$, and (6.11) is the primitive positive generator of their relation lattice. Its coefficients recover the three orbit sizes without labels. Orbit–stabilizer then gives $12/1, 12/4, 12/6$. If the vectors are replaced by primitive generators of their rays, the relation becomes $1 : 2 : 1$, which proves the final warning. $\square$

6 Modular depth

Because an H_3 sheet has eleven points in characteristic eleven, the sum of its basis vectors is simultaneously constant and augmentation-zero. Linearizing the three incidence rows therefore kills exactly this constant direction and leaves the two-dimensional profile plane. The following module statement identifies that loss.

Retain the notation of Section 5, put

$$k = \mathbb{F}_{11}, \qquad L = G^+ = \mathrm{PSL}_2(11), \qquad V = k[X_+] \simeq \mathrm{Ind}_H^L \mathbf{1},$$

and regard V as the permutation module on one eleven-point sheet. Let e_1, e_4, e_6 be the indicator functions of the three K-orbits, in orbit-size order. Thus

$$V^K = ke_1 \oplus ke_4 \oplus ke_6, \qquad \mathbf{1} = e_1 + e_4 + e_6. \tag{7.1}$$

Linearizing the incidence construction in (6.7) gives

$$\mathcal{D} : V^K \longrightarrow k^4, \qquad e_i \longmapsto i\, u_i, \tag{7.2}$$

where $u_1 = v_1$, $u_4 = v_2$, and $u_6 = v_3$ are the profiles on the orbits of the indicated sizes. Equation (6.11) says exactly that $\mathcal{D}(\mathbf{1}) = 0$.

Proposition 6.1 (Modular depth quotient)**.** *The L-module V is the projective cover $P(\mathbf{1})$ and has Loewy dimensions*

$$1 \mid 9 \mid 1.$$

The map $\mathcal{D}$ has kernel $k\mathbf{1}$ and induces an isomorphism

$$P(\mathbf{1})^K / \operatorname{soc} P(\mathbf{1}) \xrightarrow{\sim} \langle u_1, u_4, u_6 \rangle_k. \tag{7.3}$$

The sheet exchanged by J carries the negative copy of this quotient.

Proof mode. Projectivity, indecomposability, fixed-point exactness, and the socle quotient are conceptual. Absolute irreducibility of the nine-dimensional middle layer and the rank and kernel of $\mathcal{D}$ are exact generator-matrix calculations in the `relative-cubic-depth` bundle, reconstructed by its independent replay.

Proof. The order of $H \simeq A_5$ is prime to 11, so its trivial module is projective and induction makes V projective. The action of L on X_+ is doubly transitive. Hence $\mathrm{End}_L(V)$ is generated by the identity and the all-ones operator E. In characteristic eleven,

$$E^2 = 11E = 0,$$

so this endomorphism algebra is local. Thus V is indecomposable. Its trivial head occurs once, and consequently $V = P(\mathbf{1})$. The constant vector is its one-dimensional socle. The remaining nine-dimensional layer is absolutely irreducible; equivalently, the exact generator matrices on it span the full 9-by-9 matrix algebra. This gives the displayed Loewy dimensions.

Taking K-fixed points is exact because $11 \nmid |K| = 12$. The orbit sizes $1, 4, 6$ give (7.1). The three vectors in (7.2) span the plane (6.9), and their only relation is (6.11), so $\ker \mathcal{D} = k\mathbf{1}$. This proves (7.3). Finally $D(JM) = -D(M)$, which attaches the outer character on the opposite sheet. □

Corollary 6.2 (The canonical nine-space bridge)**.** *For the fixed marked conic and sheet X_+, there is a canonical G^+-module isomorphism*

$$\mathrm{rad}\, P(\mathbf{1}) / \mathrm{soc}\, P(\mathbf{1}) \xrightarrow{\sim} M_8{\downarrow_{G^+}} . \tag{7.4}$$

It sends the augmentation class of $\sum_{M \in X_+} a_M [M]$ to

$$\sum_{M \in X_+} a_M\, \pi_8(x_M), \qquad \sum_M a_M = 0, \tag{7.5}$$

where π_8 is the top-harmonic projection. Formula (7.5) is independent of the reference matching.

Proof. The unique trivial head map in Proposition 6.1 is the augmentation map, so

$$\mathrm{rad}\, P(\mathbf{1}) = \ker(V \xrightarrow{\Sigma} k).$$

Changing the reference matching translates every x_M by the same vector, so the augmentation condition makes (7.5) reference-independent. The affine cocycle term in x_{gM} cancels for the same reason, hence the map is G^+-equivariant. The same-sheet first-moment identity of Proposition 4.1 sends the constant vector $\mathbf{1}$, which lies in the augmentation module because $|X_+| = 11 = 0$ in k, to zero. Thus (7.5) descends to $\mathrm{rad}\, P(\mathbf{1}) / \mathrm{soc}\, P(\mathbf{1})$.

Its image contains a nonzero same-sheet quotient and is G^+-stable. The irreducibility of $M_8 {\downarrow_{G^+}}$ therefore makes it surjective. Both sides have dimension nine, so it is an isomorphism. □

This quotient identifies the precise modular loss in the six-profile construction. Before linearization, the six profiles remain distinct and recover all six double-coset labels. After linearization, the constant orbit trade dies and only the two-dimensional depth plane remains. Conversely, the three rational rays in that plane remain distinct but do not retain their orbit weights. The weights $1, 4, 6$ are recovered only when the quotient is supplied with the incidence-normalized vectors of (6.8), as in Corollary 5.3.

Remark 6.3 (Reduction of the profile table). The four-coordinate profiles in (6.8) are integral before reduction. The greatest common divisor of their nonzero 2-by-2 minors is 3. Their span therefore drops below rank two only in characteristic 3. The six signed rows remain distinct away from characteristics 2 and 3: characteristic 2 identifies antipodes, while characteristic 3 sends the rows v_1 and v_2 to zero. This is a statement about the displayed integral table, not a construction of the marked conic geometry in those characteristics.

7 Arithmetic splitting of the marker algebra

The arithmetic representatives of the three configurations come in conjugate pairs. Besides the base matchings in (4.1), we use

$$\begin{aligned} M_{B,-} &= \{\{0,\infty\},\{1,3\},\{2,6\},\{4,5\}\}, \\ M_{B,+} &= \{\{0,\infty\},\{1,5\},\{2,3\},\{4,6\}\}, \\ M_{H,-} &= \{\{0,10\},\{1,\infty\},\{2,7\},\{3,5\},\{4,8\},\{6,9\}\}. \end{aligned} \tag{8.1}$$

The pair for A_3 consists of two copies of M_{A_3}, the pair for B_3 is $(M_{B,-}, M_{B,+})$, and the pair for H_3 is $(M_{H_3}, M_{H,-})$. To make the arithmetic bridge part of the definition, let $\bar{k}$ be an algebraic closure of the working field and define the root-specialization maps

$$\begin{array}{ccc} T & \text{root} & \sigma_T(\text{root}) \\ \hline A_3 & \sqrt{2},\ -\sqrt{2} & M_{A_3},\ M_{A_3} \\ B_3 & 3,\ 4 & M_{B,-},\ M_{B,+} \\ H_3 & 4,\ 8 & M_{H_3},\ M_{H,-}. \end{array} \tag{8.2}$$

Thus the *marker datum* is the quadratic algebra together with σ_T, not the polynomial alone. Exact projective transport places the displayed B_3 and H_3 values in Ω_{B_3} and Ω_{H_3}. The certificate also verifies that σ_T is the reduction of the corresponding integral Coxeter frames; the theorem below needs only the explicit map (8.2). We call the datum *fused* when its two geometric roots have one rational matching value, and *split* when they give two distinct rational matching sheets.

Lemma 7.1 (Split–inert frame calculation)**.** *The three marker polynomials factor as follows:*

$$\begin{array}{rcc} A_3,\ q=5 & t^2-2 & \textit{irreducible}, \\ B_3,\ q=7 & t^2-2=(t-3)(t-4) & \textit{split}, \\ H_3,\ q=11 & t^2-t-1=(t-4)(t-8) & \textit{split}. \end{array} \tag{8.3}$$

In the first row the two geometric roots form one Frobenius orbit over $\mathbb{F}_{25}$. In each split row the nontrivial involution of the quadratic marker algebra exchanges the two rational roots.

Proof. The nonzero squares modulo 5 are 1 and 4, so 2 is not a square. If $u^2 = 2$ in $\mathbb{F}_{25}$, the nontrivial $\mathbb{F}_5$-automorphism sends u to $-u$, and hence exchanges the two roots. The remaining factorizations follow by direct substitution. □

Theorem 7.2 (Rank-three arithmetic gluing)**.** *For these arithmetic representatives of the rank-three matching configurations, the following hold.*

(a) *The A_3 marker datum is fused over $\mathbb{F}_5$. The B_3 and H_3 marker data split into two rational matching sheets over $\mathbb{F}_7$ and $\mathbb{F}_{11}$, respectively.*

(b) *In the H_3 case, the stabilizers H_+ and H_- of the singleton matching in each rational sheet are isomorphic to A_5, satisfy*

$$H_+ \cap H_- = K \simeq A_4, \qquad \langle H_+, H_- \rangle = \mathrm{PSL}_2(11), \tag{8.4}$$

and are exchanged by a projectivity J of nonsquare determinant. The normalizer $N_G(K)$ is a rational S_4 whose determinant-square kernel is K.

(c) *The determinant character that distinguishes the two rational sheets also negates the depth profiles. Thus choosing one root in the split marker algebra chooses one sheet, while its singleton profile chooses the corresponding decorated matching row. Forgetting that choice leaves the unordered pair.*

Proof mode. Lemma 7.1 and the determinant-character deductions are conceptual. The agreement of the specialization map (8.2) with the integral Coxeter frames, the matching stabilizers and intersections, and the generated subgroup in (8.4) are exhaustive exact finite checks. The classical subgroup names use the standard subgroup structure of $\mathrm{PGL}_2(11)$ [1, 4].

Proof. By definition, the two roots in the A_3 row of (8.2) have the same matching value. The two values in the B_3 row are exchanged by $x \mapsto -x$, and the two values in the H_3 row are exchanged by

$$x \longmapsto \frac{x-1}{x+1}.$$

The exact transport check verifies these identities on the displayed matchings. Together with Lemma 7.1, they prove (a).

For H_3, exact projective closure gives

$$|H_+| = |H_-| = 60, \qquad |H_+ \cap H_-| = 12, \qquad |\langle H_+, H_- \rangle| = 660.$$

Both stabilizers lie in the determinant-square subgroup, while the displayed transporter has nonsquare determinant. The normalizer $N_G(K)$ has order 24, and its determinant-square kernel has order 12. The cited subgroup identifications give (8.4) and the S_4/A_4 hinge, proving (b).

The determinant-square subgroup has the two matching sheets as its orbits, and J exchanges them. Section 5 proves $D(JM) = -D(M)$ and identifies the unique singleton in each sheet. Theorem 5.1 then returns the decorated matching row from that singleton. This proves (c). $\square$

Arithmetic splitting and modular depth retain different amounts of information. Splitting produces a two-point choice of sheet. The depth quotient is defined only after the A_4-refinement has been selected and remembers all three orbit rays modulo the constant socle. The first choice orients the second, but does not canonically identify it with any cubic quotient of the ten-dimensional factorization space. Appendix C makes that boundary precise.

Conclusion

Restriction to a conic erases the pairing of marked points, but division by its equation retains a layered record of the factorization. The four-endpoint switch produces an affine quotient configuration. For the three rank-three systems its span has dimensions $3, 6, 10$; the top harmonic part survives, as does the deepest radial line in even degree, while type H_3

loses exactly $Q\mathcal{H}_2$. For H_3, the base-choice class generates $H^1(\mathrm{PGL}_2(11), M_8)$ and defines the unique nonsplit homogeneous extension, while the canonical nine-space bridge identifies $\operatorname{rad} P(\mathbf{1})/\operatorname{soc} P(\mathbf{1})$ with $M_8 \downarrow_{\mathrm{PSL}_2(11)}$.

The nonlinear record is sharper. In the balanced B_3 and H_3 cases, quadratic products recover the unordered pair of sheets and a cubic signed moment orients them. After a chosen A_4-refinement, six incidence profiles separate the six matching rows even though their linear image is only a plane. The modular and arithmetic theorems identify, respectively, the constant direction lost on linearization and the rational choice that splits or fuses the marker algebra.

The limits are equally explicit. Balanced moments do not choose the A_4-refinement; nonsingleton profiles recover orbits rather than individual matchings; and the relative-cubic Tate plane does not canonically identify with the depth plane. The theorem is uniform in its mechanism, but the A_3 and B_3 spanning assertions still use exact row reductions. Replacing those two remaining rank calculations by parallel representation-theoretic nonvanishing arguments is the remaining step toward a fully uniform rank-three theorem; the incidence refinements remain finite exact inputs by design.

A Cubic structure of the profile plane

Lemma A.1 (Three-ray cubic forcing). *Let k have characteristic different from 2 and 3. Suppose u_1, u_2 form a basis of a two-dimensional k-space and $n_1, n_2, n_3 \in k^\times$ satisfy*

$$n_1 u_1 + n_2 u_2 + n_3 u_3 = 0.$$

For the signed antipodal measure

$$\xi = \sum_{i=1}^{3} n_i(\delta_{u_i} - \delta_{-u_i}),$$

the moments of degrees one and two vanish, while the cubic moment is nonzero.

Proof. The relation kills the first moment, and antipodality kills every even moment. Write

$$u_3 = -\frac{n_1}{n_3}u_1 - \frac{n_2}{n_3}u_2.$$

In $\sum_i n_i u_i^{\odot 3}$, the coefficient of $u_1^{\odot 2} \odot u_2$ is

$$-\frac{3n_1^2 n_2}{n_3^2},$$

which is nonzero. The signed cubic is twice this tensor and is therefore nonzero as well. □

Corollary A.2 (The mass-zero cubic flag). *In Lemma A.1, suppose also that $n_1+n_2+n_3 = 0$. Then its half-cubic factors as*

$$\frac{1}{2}M_3(\xi) = \sum_{i=1}^{3} n_i u_i^{\odot 3} = \frac{n_1 n_2}{(n_1+n_2)^2}(u_1 - u_2)^{\odot 2} \odot ((2n_1+n_2)u_1 + (n_1+2n_2)u_2). \tag{A.1}$$

The two displayed lines are distinct. Hence the cubic intrinsically defines a doubled line and a residual simple line; its Hessian recovers the doubled line.

Proof. Since $n_3 = -(n_1 + n_2)$, the weighted vector relation gives

$$u_3 = \frac{n_1 u_1 + n_2 u_2}{n_1 + n_2}.$$

Substitution and expansion give (A.1). The residual factor would equal the doubled factor only if $3(n_1 + n_2) = 0$, which is impossible under the hypotheses. The Hessian of a binary cubic L^2R, with L and R distinct, is a nonzero scalar multiple of L^2. $\square$

For the incidence-normalized profiles of Section 5, put

$$\eta = \delta_{v_1} + 4\delta_{v_2} + 6\delta_{v_3} - \delta_{-v_1} - 4\delta_{-v_2} - 6\delta_{-v_3}.$$

Then

$$M_j(\eta) = (1 - (-1)^j) \sum_{i=1}^{3} n_i v_i^{\odot j}, \qquad (n_1, n_2, n_3) = (1, 4, 6). \tag{A.2}$$

Equation (6.11) kills the first moment, and antipodality kills every even moment. The first two profiles are independent over $\mathbb{F}_{11}$, since the determinant of their first two coordinates is $-18 \neq 0$. Lemma A.1 therefore forces the cubic to be nonzero. In the frozen normalization, its first coordinate is

$$2((-6)^3 + 4(-3)^3 + 6(3)^3) = 6 \neq 0 \quad \text{in } \mathbb{F}_{11}. \tag{A.3}$$

Thus the six-row quotient preserves the cubic-first signed moment even though its linear image is only a plane.

Remark A.3 (The H_3 cubic flag). Here $1 + 4 + 6 = 0$ in $\mathbb{F}_{11}$, so Corollary A.2 forces the double-line factorization. In the basis $e_1 = v_2, e_2 = v_3$ of the profile plane, the projective cubic is

$$f(x, y) = x^3 + 7x^2y + 5xy^2 + 9y^3 = (x - y)^2(x - 2y),$$

with Hessian $7x^2 + 8xy + 7y^2 = 7(x - y)^2$. Hence the cubic intrinsically distinguishes a doubled line and a residual simple line. This flag is internal to the decorated profile plane: it supplies no canonical map from the full relative-cubic space and no descended G^+-action.

B Verification architecture

Table 1 separates the mathematical arguments from their exact finite inputs. Here "certificate" means a deterministic calculation over a prime field, checked against a canonical JSON output and a SHA-256 manifest; an independent replay uses a separate implementation. "Lean" means that the stated finite implication is checked by the Lean kernel. Neither term replaces the cited identification of the classical groups or configurations.

The claim-by-claim map is `verification/trust_manifest.json`; the exact twenty-statement identity is `verification/statement_identity.json`. From the root of the archival source, the aggregate entry point is

```
python3 papers/clebsch-factorization/verification/verify_release.py
```

It checks the statement identity and all admitted checksum manifests, runs the admitted primary checks and independent replays, elaborates the arithmetic-gluing Lean gate, and rebuilds this manuscript through the repository Makefile.

The enumerated review closure accompanying this candidate is pinned by

`verification/evidence_fingerprint.json`.

Result	Proof in the text or classical input	Exact evidence
Matching-secant quotient	unique conic divisibility and the four-endpoint switch	none
Ranks and the middle Fischer layer	harmonic decomposition; Edge–Dye configurations	quotient matrices; for H_3, the equivariant cohomology reduction and one normalized scalar; independent replay
Balanced sheets and cubic orientation	Proposition 4.1; tensor identities	exact moments; radical hypotheses and independent replay
Six-profile reconstruction	subgroup marks, orbit–stabilizer, and the three-ray lemmas	profile rows and replay; decorated-parent bijection
Modular depth and the Tate plane	projectivity, fixed-point exactness, and standard modular-module input	invariant, coinvariant, and depth calculations with replay
Arithmetic splitting and gluing	displayed root calculations; cited subgroup identifications	Lean gate and normalized certificate/replay

Table 1: Proof modes for the principal results.

Its SHA-256 digest is

```
c3645ef980ddc4a9c04a85ca5694b80fdfe811
c10989544c17e285f71f7c4f8c.
```

That fingerprint fixes a normalized manuscript, statement contents, extractor, paper Makefile, verification documentation, eight evidence manifests, command vectors, Lean gate, runner, trust manifest, and environment: Python 3.13.12, Lean 4.32.0-rc1, and Mathlib revision

```
571b8a8e54219b4d393f75f4b8653fac08197fcc.
```

A successful replay ends with

```
clebsch factorization release:  CHECK OK.
```

The normalization replaces only the displayed fingerprint digest and the statement identity's derived full-source hash, avoiding a circular self-hash. This is an explicit review-source allowlist, not a claim to hash every file in the repository or every transitive tool dependency. An immutable public archive locator must be added before release.

The largest formal component is the gate

```
RelativeConicArcs.Gates.
ClebschArithmeticGluing.
```

Its twenty-three terminals check the displayed roots, matching involutions, projective orders, determinant halves, the A_3/B_3 orbit calculations, and the split–fused conclusion. For H_3, Lean checks the normalized certificate rows, intersections, determinant classes, the $11 + 11$ signature split, and the shape of the bounded generation words. Evaluation and completeness of the 660-element coset and word certificates remain exact generator/replay obligations. The classical names S_4, A_4, A_5, and $\mathrm{PSL}_2(11)$ are cited inputs: no group name is inferred from an order.

The other computations are load-bearing only at the points shown in Table 1. The matching-module bundle supplies the three finite quotient ranks and frozen moment data. The H_3-equivariant-rank bundle checks the Sylow-normalizer weights, A_5-fixed dimensions, sheet membership, and minimal top and radial witnesses. The balanced-sheet bundle checks the finite hypotheses of the symbolic radical argument, while the Gorenstein bundle checks the ideals, deletion ranks, socles, and resolutions. The profile-incidence bundle computes one row per double coset; the decorated-parent bundle identifies a singleton matching with its parent; the relative-cubic-depth bundle computes the modular and Tate rank patterns; and the arithmetic-gluing gate checks the normalized splitting data. All use exact prime-field arithmetic. The general divisibility theorem, radical–Hadamard lemma, three-ray cubic lemma, and mass-zero factorization are proved in the text and do not depend on those enumerations.

This verification surface proves only the three marked configurations defined here. It does not supply an all-characteristic construction, a canonical map from the relative-cubic Tate plane to the depth plane, or any passage, holonomy, theta, Fourier, quantum, or Mathieu identification.

C The relative-cubic Tate plane

Let $W = W_{H_3}$ be the ten-dimensional quotient space from Theorem 3.1, let χ be the determinant character of $G = \mathrm{PGL}_2(11)$, and put

$$M = \mathrm{Sym}^3(W) \otimes \chi, \qquad R = M^G.$$

The signed cubic of Section 4 lies in R, which has dimension three. There is a canonical two-dimensional quotient of this source space, but it is not canonically the depth plane of Proposition 6.1.

Proposition C.1 (Canonical relative-cubic Tate plane)**.** *The invariant space R and the coinvariant space M_G both have dimension three. The canonical projection and group norm fit into*

$$R \xrightarrow{\pi} M_G \xrightarrow{N} R, \qquad \operatorname{rank}\pi = 2, \quad \operatorname{rank} N = 1, \tag{C.1}$$

with

$$\operatorname{im}\pi = \ker N, \qquad \operatorname{im} N = \ker\pi = \langle [1:3:9] \rangle. \tag{C.2}$$

Contraction by the covector whose square is the unique rank-one member of the invariant symmetric-form pencil gives a second construction of the same quotient. In the frozen relative basis its matrix is

$$\begin{pmatrix} 8 & 1 & 0 \\ 0 & 8 & 1 \end{pmatrix}, \tag{C.3}$$

and its kernel is again $\langle [1:3:9] \rangle$.

Proof mode. The invariant/coinvariant construction and the implications of the displayed ranks are conceptual. The dimensions, norm and projection ranks, kernel line, rank-one pencil member, contraction matrix, and flattening census are exact finite linear algebra in the `relative-cubic-depth` bundle and its independent replay.

Proof. Exact invariant and commutator calculations give $\dim R = \dim M_G = 3$, the two ranks in (C.1), and $N\pi = 0$. Thus $\operatorname{im}\pi \subseteq \ker N$; both spaces have dimension two, so they are equal. The computed image of N is the displayed kernel of π, proving (C.2).

The invariant symmetric-form pencil on W has a unique rank-one member ℓ^2, where ℓ transforms by χ. Contracting a χ-twisted cubic by ℓ cancels the two characters and lands in the invariant quadratic pencil. Exact contraction gives (C.3), whose kernel proves the second assertion. □

The kernel line is intrinsic rather than an artifact of the displayed basis. Among the 133 projective lines of R, it is the unique line whose first flattening has rank 9; the remaining census is one line of rank 1 and 131 lines of rank 10.

Remark C.2 (Why the two canonical planes do not glue). The three labelled cubic orbit classes project to M_G with relation $[2, 9, 1]$, whereas the three labelled depth profiles have relation $[2, 8, 1]$. There is therefore no label-preserving linear map carrying these source classes to the depth rays. The integral transfer makes the obstruction structural. In orbit-size order set

$$s = (1, 1, 1)^t, \qquad B = s(1, 4, 6).$$

Then $B^2 = 11B$. Divided transfer kills every integral balanced relation a with $(1, 4, 6)a = 0$, but $Bs = 11s$, so it fixes the depth socle after division by 11. It separates the source relation from the depth socle instead of identifying them. The doubled-line and residual-line flag of Remark A.3 is internal to the depth plane and supplies no missing map.

References

[1] J. H. Conway, R. T. Curtis, S. P. Norton, R. A. Parker, and R. A. Wilson, *Atlas of Finite Groups*, Clarendon Press, Oxford, 1985.

[2] W. L. Edge, Conics and orthogonal projectivities in a finite plane, *Canadian Journal of Mathematics* **8** (1956), 362–382.

[3] R. H. Dye, Hexagons, conics, A_5 and $\mathrm{PSL}_2(K)$, *Journal of the London Mathematical Society* (2) **44** (1991), 270–286. doi:10.1112/jlms/s2-44.2.270.

[4] M. Giudici, Maximal subgroups of almost simple groups with socle $\mathrm{PSL}(2, q)$, arXiv:math/0703685, 2007.

[5] D. Eisenbud and S. Popescu, The projective geometry of the Gale transform, *Journal of Algebra* **230** (2000), 127–173. doi:10.1006/jabr.1999.7940.

[6] G. Rodríguez-Pajares, D. Ruano, and F. Salizzoni, A combinatorial description of when a self-associated set of points fails to be arithmetically Gorenstein, arXiv:2512.16766v1, 2025.

[7] T. Rudd, *Deep-hole rigidity of the Clebsch hexagon code*, working paper, 2026.

[8] T. Rudd, *Local-unitary rigidity and transversal Clifford groups for MDS–CSS AME states*, working paper, 2026.

[9] T. Rudd, *Exact prescribed-hole defect and matching-design rigidity for projective arcs, with applications to arcs complete outside a conic*, companion manuscript, 2026.

[10] T. Rudd, *Deep holes of projective Reed–Solomon codes beyond redundancy four: cubic pencils and coherent polar flags*, working paper, 2026.

[11] Z. Raissi, C. Gogolin, A. Riera, and A. Acín, Constructing optimal quantum error correcting codes from absolutely maximally entangled states, arXiv:1701.03359, 2017.

[12] T. Braun and G. Nebe, Orthogonal representations of $\mathrm{SL}_2(q)$ in defining characteristic, *Beiträge zur Algebra und Geometrie* (2025). doi:10.1007/s13366-024-00763-w.